GEORGE CHRIS CARTER

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SENIOR SOFTWARE ENGINEER | FULL-STACK RUBY ON RAILS

Senior full-stack engineer with nearly five years building and running the research-administration, HR and innovation web applications used across Mass General Brigham. Grew from junior to lead to senior developer on a small team that runs close to 40 production Ruby on Rails apps, ten of them mine: took over four legacy systems, was first on the team to carry the portfolio from Rails 6 to 7 to 8, rewrote two apps (one from Ember to Rails), built three new applications and the team's Okta single-sign-on engine as sole developer, and was first to move the team's systems from on-premises hosting to AWS.

SKILLS

Languages & Frameworks: Ruby 4 / Rails 8.1 (maintained apps on 5.2 through 8.1), JavaScript (ES6), Stimulus, Hotwire/Turbo, jQuery, HAML, Sass, HTML5/CSS3, Bootstrap 4/5, esbuild, importmap, TinyMCE

Data & Background Processing: MySQL, ActiveRecord, multi-database Rails apps, Redis, Solid Queue, Delayed Job, scheduled (cron) imports, reminders and reports

Testing & Code Quality: RSpec, FactoryBot, Capybara/Selenium, SimpleCov, Guard, RuboCop, Brakeman and Cycode static security scanning, CodeClimate/qlty, HoneyBadger

Delivery & Infrastructure: AWS cloud hosting, Docker, GitHub Actions CI, Git/GitHub pull-request workflow, Capistrano deploys to Linux VMs, Puma, Rails engines/plugin architecture

Auth & Integrations: LDAP (net-ldap), Okta OIDC SSO (PKCE, JWKS), Microsoft Graph API, IdM and HR (PeopleSoft, Workday) data syncs, REST API design with token auth, Faraday, ActionMailer digests, PDF (wicked_pdf), Excel/CSV (caxlsx, roo), Chart.js

EXPERIENCE

Mass General Brigham – MGB Digital Resource Operations, Research Informatics Core | Boston, MA **Jan 2026 – Present**

Massachusetts General Hospital – Center for Systems Biology, Research Informatics Core | Boston, MA **Dec 2021 – Jan 2026**

Same team throughout; the Research Informatics Core moved from MGH into Mass General Brigham in January 2026.

Systems Engineer II – Senior Developer, Mass General Brigham **Jan 2026 – Present**

- Designed and built the Office for Clinical Research (OCR) protocol review platform solo (go-live Dec 2025): versioned templates, multi-stage review workflow with auto-assignment, comment release with digest emails, institution routing; ~400 protocols processed in ten months.
- Integrated OCR with the BWH CCI and MGH TCRC systems via a push-model REST API (token auth, Faraday with retries, async sync job) plus receiving-side processors; every approved protocol has synced through it since go-live.
- Built the Office of Animal Welfare Assurance (IACUC) inspection platform from scratch (Mar 2026): inspection cycles, findings and corrective actions, escalation rules, templated email with distribution lists, Insight facility import; 140+ RSpec files.
- Wrote the team's Okta OIDC single-sign-on Rails engine (authorization code with PKCE, JWKS token verification, outage fallback, RP-initiated logout, cutover preflight report) and folded it into the shared auth engine used by every app.
- First on the team to migrate my applications from the on-premises DIPR hosting environment to the AWS cloud, containerizing them with Docker and standing up GitHub Actions CI (RSpec, RuboCop, Brakeman, dependency audits).
- First on the team to move the portfolio to Ruby 4 / Rails 8.1 across seven production apps, then ran security and performance passes on each (N+1 fixes, Brakeman/Cycode findings, Dart Sass migration, dead-code removal).

Software Engineer – Lead Developer, Massachusetts General Hospital **Dec 2023 – Jan 2026**

- Led the rewrite of LabSec (lab safety and compliance) from an Ember front end into one Rails app: inspection surveys and events, corrective actions and reminders, nightly Insight data sync, Pundit/Guardian permissions; 700+ commits.
- Started the OCR platform (Aug 2025): designed the versioned-template and review-workflow data model and shipped the submission, review and admin screens that went live in Dec 2025.
- First on the team to move the portfolio from Rails 6 to 7, upgrading Evals, EDC and the suggestion apps to Rails 7.1 / Ruby 3.3 and clearing deprecations ahead of the Rails 8 upgrade.
- Primary developer on Evals, the enterprise performance-evaluation system (~92,000 evaluations processed during my tenure): employee search, a rewritten bulk employee-record importer and IdM sync, optimized notification

queries, and the PeopleSoft-to-Workday transition.
- Rewrote iSuggest, the enterprise suggestion platform (notification preferences, reminder schedules, role consolidation, status workflow); migrated the suggestion-triage app to a new VM with a test-coverage push; replaced Froala with TinyMCE on EDC.

Software Engineer – Junior Developer, Massachusetts General Hospital **Dec 2021 – Dec 2023**

- Took over four production Rails apps (Evals, iSuggest, EDC and the suggestion triage app) and shipped tracked features through code-reviewed pull requests from the first week: leaderboards, comments, status history, session timeout, suggestion mail to all users.
- Owned Evals workflow features: second-evaluator rules, reporting-structure violation notifications, institution default templates, statistics CSV exports and dashboards, inactivation rules, duplicate-record cleanup, template usage reporting.
- Made EDC configurable per contest (PFAC and Primary Care): contest-specific settings and validation, customizable submission and application pages, text-field limits, password requirements, HoneyBadger error reporting.
- Carried the first framework upgrades on the legacy apps (Rails 5.1 / Ruby 2.6, then Rails 6.0.6.1 on Evals; Rails 6.0.5.1 security patch) and added RSpec/FactoryBot coverage with each change.
- Built my first application solo from inception, the NHPCSG nonhuman-primate consortium grant portal (May 2023): funding rounds, application forms, review scoring, steering-committee access, Excel/PDF exports, external user accounts.

APPLICATIONS BUILT AND MAINTAINED AT MGB

- **OCR – Office for Clinical Research** – protocol submission and multi-stage review platform; sole developer, Rails 8.1 (2025–present).
- **OAWA – Office of Animal Welfare Assurance** – IACUC facility inspection, findings and corrective-action tracking; sole developer, Rails 8.1 (2026–present).
- **LabSec** – lab safety inspections, surveys, action items and compliance reporting; led the Ember-to-Rails rewrite (2025–present).
- **NHPCSG** – nonhuman-primate consortium grant application, review and funding-round portal; sole developer (2023–present).
- **Evals** – enterprise employee performance-evaluation system with HR data sync; primary developer (2022–present).
- **iSuggest** – MGB staff suggestion platform with Microsoft Graph inbox integration; rewrote and maintain (2022–present).
- **EDC – Ether Dome Challenge** – innovation-contest submission, judging and finalist site; maintainer (2022–present).
- **Staff suggestion triage app** – themes, voting, department routing and status workflow; maintainer (2021–present).
- **CCI and TCRC** – BWH and MGH clinical research center systems; built the OCR intake API and fund-linking processors (2026).
- **Shared Rails engines** – okta_sso (sole author), site (auth, groups, permissions), cmir_upload, search_options and other internal plugins used by every app.

EDUCATION

Actualize Coding Bootcamp – Certificate in Full Stack Web Development Jan 2021 – May 2021
Keene State College – Bachelor of Arts, Psychology 2013

PRIOR EXPERIENCE

Theatrical scenic and properties leadership, 2013 – 2021: Lead Scenic, The Public Theater (2017 – 2021); Properties Coordinator, The New School College of Performing Arts (2015 – 2017); Head of Properties, Theatre for a New Audience (2013 – 2016); Scenic Charge, Weathervane Theatre (2013 – 2016). Managed builds, budgets and crews on fixed production deadlines.